

Age-Stratified SEIR Model with Correlated Heterogeneity

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Overview

This model (SEIR_pde.py) implements SEIR dynamics with correlated individual-level heterogeneity in susceptibility (θ) and infectiousness (ϕ). The correlation structure captures the biological reality that individuals with higher susceptibility tend to exhibit higher infectiousness upon infection. The implementation solves a system of integro-differential equations using finite-dimensional discretization.

Model Features

Population structure: - Age-stratified model with 16 age groups (0-1, 1-2, ..., 14-15, 15+) - Aging dynamics with rate $1/365$ per day between consecutive age bins - $N = 186,763$, discretized into 50 bins for θ and ϕ - 3-year simulation horizon with weekly output

Correlation method: Uses Gaussian kernel approximation:

$$P(\phi_j|\theta_i) \propto \exp\left(-\frac{(i-j)^2}{2w^2}\right)$$

The kernel width w is optimized via scalar minimization to achieve target correlation $\rho = 0.8$. This method:
- Avoids expensive sampling procedures - Guarantees stochastic matrix properties (non-negative, rows sum to 1) - Scales efficiently to large n_bins - Produces smooth conditional distributions

Age structure: The model tracks (S, E, I, R) compartments for each age group with aging flux terms that transfer individuals between adjacent age bins at rate $r_a = 1/365 \text{ day}^{-1}$ (no outflux from terminal bin).

Mathematical Formulation

Continuous PDE System

The discretized ODE system approximates the following integro-PDE:

Susceptible dynamics:

$$\frac{\partial S(t, \theta, a)}{\partial t} + \frac{\partial[r(a)S(t, \theta, a)]}{\partial a} = -\theta\lambda(t)S(t, \theta, a)$$

Exposed dynamics:

$$\frac{\partial E(t, \phi, a)}{\partial t} + \frac{\partial[r(a)E(t, \phi, a)]}{\partial a} = \int_0^\infty P(\phi|\theta)\theta\lambda(t)S(t, \theta, a)d\theta - \sigma E(t, \phi, a)$$

Infectious dynamics:

$$\frac{\partial I(t, \phi, a)}{\partial t} + \frac{\partial[r(a)I(t, \phi, a)]}{\partial a} = \sigma E(t, \phi, a) - \gamma I(t, \phi, a)$$

Recovered dynamics:

$$\frac{\partial R(t, a)}{\partial t} + \frac{\partial [r(a)R(t, a)]}{\partial a} = \gamma \int_0^\infty I(t, \phi, a) d\phi$$

Force of infection:

$$\lambda(t) = \frac{\beta}{N} \int_0^\infty \int_A \phi I(t, \phi, a) d\phi da$$

where: - $\theta \in [0, \infty)$: relative susceptibility - $\phi \in [0, \infty)$: relative infectiousness - a : age - $r(a)$: aging rate (individuals move from age a to $a + da$) - $P(\phi|\theta)$: conditional probability of infectiousness given susceptibility - β : base transmission rate - σ : progression rate ($E \rightarrow I$) - γ : recovery rate ($I \rightarrow R$) - N : total population size

Discretization Scheme

Spatial discretization: - θ discretized into n_bins bins: $\theta_i, i = 1, \dots, n_bins$ (evaluated at bin centers) - ϕ discretized into n_bins bins: $\phi_j, j = 1, \dots, n_bins$ - Age discretized into n_age bins: $a_k, k = 1, \dots, n_age$

State variables: - $S(t, \theta_i, a_k) \rightarrow S_{i,k}(t)$ - $E(t, \phi_j, a_k) \rightarrow E_{j,k}(t)$ - $I(t, \phi_j, a_k) \rightarrow I_{j,k}(t)$ - $R(t, a_k) \rightarrow R_k(t)$

Finite-dimensional ODE system:

$$\frac{dS_{i,k}}{dt} = -\theta_i \lambda(t) S_{i,k} + [\text{aging flux}]_k$$

$$\frac{dE_{j,k}}{dt} = \sum_i P_{ji} \theta_i \lambda(t) S_{i,k} - \sigma E_{j,k} + [\text{aging flux}]_k$$

$$\frac{dI_{j,k}}{dt} = \sigma E_{j,k} - \gamma I_{j,k} + [\text{aging flux}]_k$$

$$\frac{dR_k}{dt} = \gamma \sum_j I_{j,k} + [\text{aging flux}]_k$$

$$\lambda(t) = \frac{\beta}{N} \sum_k \sum_j \phi_j I_{j,k} \Delta\phi$$

Aging flux (upwind scheme):

$$[\text{aging flux}]_k = r_{k-1} X_{k-1} - r_k X_k$$

where $X \in \{S, E, I, R\}$ and $r_{n_age} = 0$ (absorbing boundary).

Integration scheme: The integral $\int I(t, \phi) d\phi$ is approximated by Riemann sum: $\sum_j I_j \Delta\phi$, where $\Delta\phi$ is the bin width.

Conservation Properties

The model preserves: 1. **Infection conservation:** Total new infections in $S(\theta)$ space equals total new exposures in $E(\phi)$ space (enforced by stochastic P matrix) 2. **Population conservation:** $\frac{d}{dt}(S+E+I+R) = 0$ (up to numerical error) 3. **Non-negativity:** All compartments remain ≥ 0 (guaranteed by non-negative initial conditions and model structure)

Implementation Notes

Correlation Matrix Construction

The Gaussian kernel ansatz assumes $P(\phi|\theta)$ has support concentrated near $\phi \approx \theta$ (in bin-index space). The optimization procedure:

1. Generates n_{samples} random θ indices
2. For candidate width w , samples ϕ indices according to $P(\phi_j|\theta_i)$
3. Computes empirical correlation between θ_{samples} and ϕ_{samples}
4. Minimizes $|\rho_{\text{empirical}} - \rho_{\text{target}}|$ using bounded scalar minimization

This produces a correlation structure that is approximately correct for the chosen marginal distributions while maintaining computational tractability.

Parameter Choices

Marginal distributions: - Susceptibility: $\text{LogNormal}(\mu = 1.0, \sigma^2 = 4.0)$ - Infectiousness: $\text{Gamma}(\text{shape} = 1.0, \text{scale} = \beta/24)$

Epidemiological parameters: - $\beta = 10.0$ (transmission rate) - $\sigma = 1/3 \text{ day}^{-1}$ (incubation period: 3 days) - $\gamma = 1/24 \text{ day}^{-1}$ (infectious period: 24 days)

These values produce baseline $R_0 \approx \beta/\gamma = 240$ in a homogeneous population, though heterogeneity reduces effective R_0 .

Computational Complexity

Operation	Complexity
P matrix construction	$O(n_{\text{bins}}^2 \times n_{\text{opt_iter}}) \approx 10^4$
ODE evaluation	$O(n_{\text{age}} \times n_{\text{bins}}^2)$
Memory	$O(n_{\text{age}} \times n_{\text{bins}}^2)$

The model scales efficiently to $n_{\text{bins}} = 50\text{-}100$ and $n_{\text{age}} = 16$.

Model Validation

The implementation includes an assertion checking infection conservation:

```
assert np.isclose(new_E_phi.sum(), new_infections_theta.sum())
```

This ensures the correlation structure does not introduce numerical artifacts during the $S \rightarrow E$ transition.